

Evaluating an Offline Detector Against Adversarial Prompt Attacks

github.com/poojakira/llm-redteam-framework

Research Brief - Poster 05

Evidence status: Refreshed against current code snapshot
08b40538c00146981d450cad7af5cedd87b887b6 and successful CI run 36783655822 on 2026-09-30.

Repository

github.com/poojakira/llm-redteam-framework - public, default branch main.

Academic Project Title

Evaluating an Offline Detector Against Adversarial Prompt Attacks

Subtitle

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

One-Sentence Contribution

A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector’s generalization gap explicit instead of hiding it behind a random split.

Method

1. Generate adversarial and benign prompt corpora across defined attack categories.
2. Train/evaluate the offline TF-IDF + Logistic Regression detector.
3. Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.
4. Emit JSON/SARIF evidence for CI.
5. Keep live-model behavior claims out of scope unless separately measured.

Current Verified Evidence

Current-main Python 3.12 CI reports:

- **175 passed, 1 skipped.**
- **94.30% statement coverage**; CI gate is 90%.
- Current benchmark values remain pinned:
 - random split F1 **1.00**
 - grouped-template F1 **0.9714**
 - novel-phrasing OOD F1 **0.7188**
 - novel-phrasing precision **0.5897**
 - novel-phrasing recall **0.92**
 - novel-phrasing false-positive rate **64% (16/25)**
- Lint, type checking, dependency audit, performance gate, external validation, and comparable OOD benchmark jobs completed successfully.

Important Negative Result

The novel-phrasing result is the key generalization check: F1 falls from **0.9714** on grouped templates to **0.7188** on OOD paraphrases, while false positives rise sharply. This is evidence of lexical/template dependence, not a production-grade semantic detector.

Limitations

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.

Reproducibility

```
git clone https://github.com/poojakira/llm-redteam-framework.git
cd llm-redteam-framework

git checkout 08b40538c00146981d450cad7af5cedd87b887b6

python -m pip install -e ".[dev]"

pytest tests/ -q --cov=redteam --cov-report=term

python benchmarks/ood_novel_phrasings.py
```

Expected CI evidence: **175 passed, 1 skipped, 94.30% coverage**.